

Department of Statistics and Business Analytics
STAT 482 - Capstone in Analytics for Business

Spring 2026

Case Title: How Much Should We Stock? Forecasting Product Demand for Inventory Planning

Due submission date: Wednesday, February 23, 2026

Case Description

Retail firms must balance product availability against the high costs of overstocking and stockouts. Accurate sales forecasting plays a critical role in inventory planning, production scheduling, and cash-flow management.

In this case study, students assume the role of data analysts for a mid-size retail company that sells a single product category nationwide. Management needs reliable short-term demand forecasts to determine appropriate inventory levels for the next few months. Historical monthly sales data exhibit both trend and seasonality, making time series analysis a suitable forecasting approach.

Students will analyze historical sales data, model underlying patterns, generate forecasts, and evaluate forecast accuracy. The final recommendation must clearly communicate forecast results and uncertainty to non-technical stakeholders.

Learning Objectives

By completing this case study, students will be able to:

- Identify and model trend and seasonal components in sales data
- Apply time series forecasting techniques for short-term demand prediction
- Evaluate forecast performance using quantitative accuracy metrics
- Interpret and communicate forecast uncertainty to decision-makers
- Translate analytical results into actionable inventory recommendations

Data Source

Primary Dataset:

Monthly sales data derived from a retail demand forecasting dataset.

- **Kaggle:** Store Item Demand Forecasting Challenge
- **Download link:**
<https://www.kaggle.com/competitions/demand-forecasting-kernels-only/data>

Students may aggregate daily data into monthly sales totals or use a provided simulated monthly dataset based on realistic retail demand patterns.

Suggested Solution Methodology (Using R)

Students are expected to conduct the analysis using R, following these steps:

1. **Data Preparation**
 - Import and clean sales data
 - Aggregate sales to monthly frequency (if required)

- Visualize time series
 - 2. **Exploratory Time Series Analysis**
 - Identify trend and seasonality
 - Perform time series decomposition (additive or multiplicative)
 - 3. **Model Development**
 - Fit forecasting models such as:
 - ARIMA
 - Exponential Smoothing (ETS)
 - Justify model selection
 - 4. **Forecasting**
 - Generate short-term forecasts (e.g., 3–6 months)
 - Include prediction intervals
 - 5. **Model Evaluation**
 - Calculate forecast accuracy metrics:
 - Mean Absolute Percentage Error (MAPE)
 - Root Mean Squared Error (RMSE)
 - 6. **Managerial Interpretation**
 - Translate forecasts into inventory planning insights
 - Discuss risks and uncertainty
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Required Deliverables

1. Written Report (8–12 pages, excluding appendix)

The written report should include:

- Executive summary
- Business context and problem definition
- Methodology:
 - Description techniques and assumptions
 - Model description and justifications
- Results:
 - Exploratory analysis and visualizations
 - Forecast results and accuracy evaluation
- Managerial implications and recommendations
- Limitations and future improvements

The report should be prepared using **R Markdown or Quarto** to ensure reproducibility.

2. Oral Presentation

- **Slides:** 8–10 slides
- **Duration:** Approximately 10 minutes (plus Q&A, if applicable)

he presentation should concisely cover:

- Business problem and objective
- Data overview
- Methodology summary
- Key findings and forecasts
- Forecast uncertainty
- Inventory recommendations